

Gauge field singularities and noninteger topological charge^{a)}

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We study the behavior of singular gauge field configurations under gauge transformations and we determine the relation between noninteger topological charge and the possibility of displacing singularities.

I. INTRODUCTION

Since the introduction of the first instanton solution,¹ many important physical effects have been attributed to its existence.²⁻⁴

At present, big efforts are being made in order to establish a theory of strong interactions based in quantum chromodynamics with a correct account of instanton effects.⁵

However, if only instanton contributions are considered in nonperturbative computations, important problems remain unsolved, namely those related to quark confinement. For this reason it has been stressed that other singular configurations may play a fundamental role.⁶

More recently there have been other suggestions emphasizing the relevance of configurations with noninteger topological charge in the quantization of non-Abelian theories.⁷⁻⁹

As it was stressed by Witten⁹ these fractional charge configurations maintain the principal characteristics of the instanton picture: existence of an infinite number of degenerate vacua, possibility of resolution of the U(1) problem. At the same time, however, they may overcome its difficulties, in particular, those which arise from the conflict between the boundary condition of pure gauge field at infinity (which leads to instantons) and quark confinement.

Motivated by these remarks, we study in this paper the possible singularities of gauge fields in order to understand their behavior under gauge transformations. Then we will be able to establish our principal results stressing the relation between noninteger topological charge and the possibility of displacing gauge field singularities.

In Sec. II we discuss some interesting Abelian and non-Abelian examples which illuminate our results and which are established in Sec. III.

II. ABELIAN AND NON-ABELIAN EXAMPLES

We shall discuss in this section some examples of gauge field configurations having peculiar properties: both in the Abelian and in the non-Abelian cases they present singularities that one cannot displace by gauge transformations. Another characteristic of the configurations is that their "local" topological charge—to be defined below—is noninteger.

The first example, an Abelian one, resembles to the c -

instantons introduced by Nielsen and Schroer in the context of the Schwinger model.¹⁰ In the non-Abelian case the configuration we discuss is singular and it behaves at the singularity not as a pure gauge but as an "almost pure gauge".¹¹ It has nonzero but noninteger local topological charge and, as we will see in Sec. III, this fact is closely related to the gauge invariance of the singularities.

A. The Abelian case

Let us consider an Abelian gauge theory with gauge group U(1) in (Euclidean) two-dimensional space. We shall study a configuration of the form

$$A_\mu^N = \sum_{i=1}^N g_i(x) \epsilon_{\mu\nu} \frac{(x^\nu - a^{i\nu})}{(x^\nu - a^{i\nu})^2} \quad (2.1)$$

with

$$g_i(x) = O(|x|^{-2}), \quad (2.2)$$

$$g_i(x) = 1 + (x_\nu - a_\nu^i). \quad (2.3)$$

Because of conditions (2.2) and (2.3), A_μ^N is singular at the points a_μ^i ($a_\mu^i \in \mathbb{R}^2$) and well-behaved at infinity.

We shall now compute the topological charge of A_μ^N from the expression

$$q[A_\mu^N] = \frac{1}{4\pi} \int_{\mathbb{R}^2} \epsilon_{\mu\nu} F^{\mu\nu} d^2x = \frac{1}{2\pi} \int_{\mathbb{R}^2} \epsilon_{\mu\nu} \partial^\nu A^\mu d^2x. \quad (2.4)$$

Since A_μ^N is $\mathbb{C}^\infty$ (excepting at points $x_\mu = a_\mu^i$), we can apply Stokes' theorem and write expression (2.5) in the form

$$q[A_\mu^N] = \lim_{R \rightarrow \infty} \frac{1}{2\pi} \oint_{S_R} A_\mu^N dx^\mu - \sum_{i=1}^N \lim_{\epsilon_i \rightarrow 0} \frac{1}{2\pi} \oint_{S_{\epsilon_i}} A_\mu^N dx^\mu, \quad (2.5)$$

where S_R is a circle of radius R and S_{ϵ_i} are small circles of radius ϵ_i surrounding each singularity. Then,

$$q[A_\mu^N] = \sum_{i=1}^N \lim_{\epsilon_i \rightarrow 0} \frac{1}{2\pi} \oint g_i d\theta^i = N. \quad (2.6)$$

Now, near the singularities, A_μ^N behaves like a pure gauge:

$$A_\mu^N \simeq \partial_\mu \tan^{-1} \left(\frac{x_2 - a_2^i}{x_1 - a_1^i} \right), \quad x_\mu \sim a_\mu^i. \quad (2.7)$$

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Hence, the singularity can be displaced by a suitable gauge transformation. For example, for the $N = 1$ case, it is evident that the configuration

$$A'_\mu = A_\mu^{N=1} - \partial_\mu \tan^{-1} \left(\frac{x_2 - a_2^1}{x_1 - a_1^1} \right) \quad (2.8)$$

gauge-transformed of $A_\mu^{N=1}$ has the singularity removed to infinity, since we can write A'_μ in the form

$$A'_\mu = [g_1(x) - 1] \epsilon_{\mu\nu} \frac{(x_\nu - a_\nu^1)}{(x_\eta - a_\eta^1)^2}, \quad (2.9)$$

regular at $x_\mu = a_\mu^1$.

For general N one applies a chain of gauge transformations of the type (2.8), finally obtaining a regular configuration of the form

$$A'_\mu = \sum_{i=1}^N [g_i(x) - 1] \epsilon_{\mu\nu} \frac{(x_\nu - a_\nu^i)}{(x_\eta - a_\eta^i)^2}. \quad (2.10)$$

Of course

$$q[A'_\mu] = q[A_\mu^N] = N. \quad (2.11)$$

One can recognize in this procedure the method developed by Giambiagi and Rothe¹² to displace the singularities of 't Hooft's multi-instanton solution.¹³ Of course, since our example is Abelian, the prescription in this case is evident.

However, if instead of configuration (2.1) we consider a gauge field of the form

$$A_\mu^C = C A_\mu^N = C \sum_{i=1}^N g_i(x) \frac{(x_\nu - a_\nu^i)}{(x_\eta - a_\eta^i)^2} \epsilon_{\mu\nu} \quad (2.12)$$

with g_i defined as in Eqs. (2.2) and (2.3), and we try to displace the singularities as we did before, then we are faced with a problem. The singular gauge transformation that we must perform is of the form

$$A_\mu^{C'} = A_\mu^C - \sum_{i=1}^N \partial_\mu \left[C \tan^{-1} \left(\frac{x_2^i - a_2^i}{x_1^i - a_1^i} \right) \right]. \quad (2.13)$$

For simplicity, let us consider the case $N = 1$, $a_\mu^1 = 0$. We see that the singular gauge transformation corresponds to the $U(1)$ element

$$g = \exp(iC\theta), \quad \theta = \tan^{-1}(x_2/x_1) \quad (2.14)$$

which is multivalued whenever $C \notin \mathbb{Z}$. If one excludes this kind of transformations, it is not possible to gauge away the singularities as we did before.

The topological charge of this configuration is

$$q[A_\mu^C] = CN \quad (2.15)$$

which in general is noninteger. Gauge fields similar to those given by Eq. (2.12) were already studied by Nielsen and Schroer in the context of the Schwinger model.¹⁰ These authors found that configurations with $q = \frac{1}{2}$, the c -instantons, saturate certain functional integrals which show confinement in the Schwinger model.

More recently Witten⁹ discussed field theories with noninteger topological charge in the context of the quark confinement problem. In fact, the first model he studied⁹ corresponds to the Lagrangian density (in two-dimensions)

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + D_\mu \phi^* D^\mu \phi - M^2 |\phi|^2 \quad (2.16)$$

with

$$\phi = \phi_1 + i\phi_2 \quad (2.17)$$

and

$$D_\mu = \partial_\mu - iA_\mu. \quad (2.18)$$

Finite action requires $|\phi|_{|x| \rightarrow \infty} = 0$ and

$$A_\mu \Big|_{|x| \rightarrow \infty} = \partial_\mu \alpha. \quad (2.19)$$

These conditions allow for configurations of the kind (2.12). It is only when $-M^2 |\phi|^2$ is replaced by a symmetry breaking potential

$$V(\phi) = -\lambda (|\phi|^2 - a^2)^2 \quad (2.20)$$

that the topological charge must be integer since condition (2.19) is replaced by:

$$|\phi|^2 \Big|_{|x| \rightarrow \infty} = a^2, \quad D_\mu \phi \Big|_{|x| \rightarrow \infty} = 0, \quad (2.21)$$

thus forbidding $C \notin \mathbb{Z}$, since the contrary is only possible for multivalued ϕ .

B. The non-Abelian case

In analogy with the Abelian case, we will try to gauge away singularities of a particular gauge field configuration A_μ^C , for the non-Abelian case, with gauge group $SU(2)$ in (Euclidean) four-dimensional space.

Let A_μ^C , be

$$A_\mu^C = \sum_{i=1}^N C f_i g_i^{-1} \partial_\mu g_i, \quad (2.22)$$

with

$$f_i = \frac{\lambda_i^2}{x^2}, \quad f_i = 1 - y_i^2/\lambda_i^2, \quad (2.23)$$

$$y_\mu^i = x_\mu - a_\mu^i, \quad \mu = 1, 2, 3, 4,$$

$$g_i = y_4^i - i \frac{\vec{\sigma} \cdot \vec{y}^i}{(y_i)^2}. \quad (2.24)$$

The configuration is singular at points $a_\mu^i \in \mathbb{R}^4$, $i = 1, 2, \dots, N$. For $C = 1$, expression (2.22) corresponds to 't Hooft's multi-instanton solution¹³ and since at points a_μ^i

$$A_\mu^{C=1} \sim_{x \rightarrow a^i} g_i^{-1} \partial_\mu g_i, \quad (2.25)$$

these singularities can be gauged away, as it was shown by Giambiagi and Rothe.¹²

Now, if $C \neq 1$, the problem is completely different: A_μ^C is no more a pure gauge at the singularities and we will see that it is not possible to gauge away them as it was done in the instanton case.

For simplicity let us consider the case $N = 1$, $a_\mu^1 = 0$. If the singularity was gauge movable, there must exist a gauge transformation h such that

$$A_\mu^C = h^{-1} A_\mu^{\text{reg}} h + h^{-1} \partial_\mu h \quad (2.26)$$

and

$$F_{\mu\nu}^C = h^{-1} F_{\mu\nu}^{\text{reg}} h, \quad (2.27)$$

where A_μ^{reg} is the gauge transformed of A_μ^C , regular at $x_\mu = 0$.

Now,

$$F_{\mu\nu}^C = \frac{4iC(1-C)}{x^4} (x_\mu \sigma_{\nu\rho} x_\rho - x_\nu \sigma_{\mu\rho} x_\rho + x^2 \sigma_{\mu\nu}). \quad (2.28)$$

(Here, $\sigma_{ij} = \frac{1}{2}\epsilon_{ijk}\sigma_k$, $\sigma_{i4} = \frac{1}{2}\sigma_i$.) It is easy to see now that $\text{tr}(F_{\mu\nu}^C F_{\mu\nu}^C)$ is singular for $C \neq 0, 1$ at $x = 0$. But since

$$\text{tr}(F_{\mu\nu}^C F_{\mu\nu}^C) = \text{tr}(F_{\mu\nu}^{\text{reg}} F_{\mu\nu}^{\text{reg}}) \quad (2.29)$$

we have the l.h.s. singular (excepting for $C \neq 0, 1$) and the r.h.s. regular. Then, Eq. (2.27) cannot be satisfied. That is there is no gauge transformation which makes A_μ^C regular at $x = 0$ for $C \neq 0, 1$.

We can compute the topological charge of A_μ^C from the expression

$$q = \frac{1}{32\pi^2} \text{tr} \int d^4x F_{\mu\nu}^* F_{\mu\nu}. \quad (2.30)$$

we will use the relation¹⁴

$$\partial_\mu K^\mu = \text{tr} F_{\mu\nu}^* F_{\mu\nu} \quad (2.31)$$

with

$$K_\mu = 4\epsilon_{\mu\alpha\beta\gamma} \text{tr}(\frac{1}{2}A_\alpha \partial_\beta A_\gamma + \frac{1}{3}A_\alpha A_\beta A_\gamma) \quad (2.32)$$

which for regular configurations allows us to write

$$q = \frac{1}{32\pi^2} \lim_{R \rightarrow \infty} \int_{\sigma_R} d^3x K_\mu n^\mu, \quad (2.33)$$

where σ_R is a surface enclosing a sphere of radius R and n^μ is its outward normal. Of course if A_μ is not regular inside σ_R , expression (2.33) is no longer true. One has to surround each singularity with a small sphere and proceed in analogy with what was done in the Abelian case [see Eq. (2.5)].

It is easy to see that near the singularities

$$K_\mu = \epsilon_{\mu\alpha\beta\gamma} 2C^2(2C/3 - 1) \text{tr}(g_i^{-1} \partial_\alpha g_i g_i^{-1} \partial_\beta g_i g_i^{-1} \partial_\gamma g_i)$$

thus giving

$$q[A_\mu^C] = C^2(3 - 2C) \sum_{i=1}^N \lim_{\epsilon \rightarrow 0} \frac{1}{24\pi^2} \times \int_{\sigma_i} \epsilon_{\mu\alpha\beta\gamma} g_i^{-1} \partial_\alpha g_i g_i^{-1} \partial_\beta g_i g_i^{-1} \partial_\gamma g_i n^\mu d^3x. \quad (2.34)$$

(There is no contribution of the integral on σ_R since $K^\mu \rightarrow 0$ sufficiently fast for $R \rightarrow \infty$.) Then

$$q[A_\mu^C] = C^2(3 - 2C) \frac{1}{24\pi^2} \sum_{i=1}^N \int_{\sigma_i} dg, \quad (2.35)$$

where dg is the invariant volume on the group. Finally, we have

$$q[A_\mu^C] = C^2(3 - 2C)N. \quad (2.36)$$

In general q is noninteger for $C \notin \mathbb{Z}$. In particular, for $C = \frac{1}{2}$ and $N = 1$ we obtain $q = \frac{1}{2}$. This case is in fact related to the meron¹⁵ since:

$$A_\mu^{\text{meron}} = \frac{1}{2} g_i^{-1} \partial_\mu g_i. \quad (2.37)$$

Of course, expression (2.37) is singular both at the origin and at infinity and then formula (2.34) gives in this case the "local" charge at the origin since it does not take into account the contribution at infinity.

Finally, we want to point out that the action of configurations of the type (2.22) is infinite. This is due to the fact that the (gauge invariant) singularities of A_μ^C render the action density not locally integrable for $C = 0, 1$. In fact, Marino and Swieca¹⁶ have already considered a large class of configurations carrying noninteger topological charge and they proved that they have necessarily infinite action.

From the examples discussed above we see that singularities and noninteger topological charges are closely related. In the next section we will try to understand more deeply the nature of the relation since, as it has been suggested, these types of configurations may play a central role in the confinement mechanism.

III. GAUGE FIELD SINGULARITIES

In this section we shall clarify the peculiar characteristics of the examples discussed before. To this end we will establish a classification of gauge field singularities which will allow us to understand their behavior under gauge transformations.

We then discuss some topological properties of singular gauge configurations and stress the relation between noninteger topological charges and the possibility of displacing an isolate singularity by a gauge transformation.

We will consider a gauge field A_μ on $\mathbb{R}^n$ with gauge group G [For the cases of interest we will take $G = U(1)$ and $SU(2)$.]

We will say that A_μ is regular at $x = x_0$ if it is C^∞ in a neighborhood of x_0 . We can consider that A_μ is defined on S^n ¹⁷ with an eventual singularity at the point at infinity. In general A_μ can also present singularities on $\mathbb{R}^n$. We will call S the set of all possible singularities of A_μ .

The behavior of A_μ near its singularities determines whether or not it defines a connection on a principal fibre bundle $P(S^n, G)$ with base manifold S^n . We recall that in the framework of the fibre bundle theory, a gauge transformation corresponds to a change of trivialization of $P(S^n, G)$.¹⁸

We will distinguish two possible types of singularities of A_μ at $x_0 \in S^n$:

(i) *Gauge-movable (GM) singularities*: Those singularities which allow the definition of a connection on a principal fibre bundle $P(S^n, G)$. These type of singularities are produced by the particular trivialization in which one is considering the connection. Then, they can be removed by a gauge transformation.

(ii) *Gauge-invariant (GI) singularities*: Those singularities which prevent the definition of a connection on a principal fibre bundle $P(S^n, G)$ from A_μ . That is, the singularities that are not evitable by a gauge transformation.

If one is faced with a GM singularity of A_μ at x_0 , one can obtain an A'_μ , gauge-transformed of A_μ , regular at x_0 , by a suitable singular gauge transformation. An example of this kind is the 't Hooft multi-instanton configuration¹³ [given by expression (2.22) with $C = 1$] with singularities at $S = \{a'_i, i = 1, 2, \dots, N\}$. In fact, in Ref. 12 it is given the explicit gauge transformation, singular at points of S , which sends all the singularities to infinity, thus giving a multi-instanton configuration regular on $\mathbb{R}^n$.

In the fibre bundle language, this corresponds to passing from a trivialization defined on $S^n - S$ to another one defined on $S^n - \{ \infty \}$

Of course this procedure is also applicable to the Abelian example defined by Eq. (2.1), since it also presents a GM singularity.

On the other hand, if the singularity is gauge-invariant, it cannot be removed by a change of trivialization. To displace such a singularity it will be necessary to make another kind of transformation associated with other possible invariances of the field theory. An example of this kind is provided by the meron configuration¹⁵ [Eq. (2.38)] which is singular both at the origin and at infinity. The singularities can be displaced only by a conformal transformation to two distinct points $u \neq v$. In general this kind of transformations prevent the displacement of two singularities to the same point.

The Abelian configuration (2.12) and the non-Abelian one, [Eq. (2.22) with $C \notin \mathbb{Z}$ also present GI singularities.

It is well known that a principal fibre bundle $P(S^n, G)$ is topologically characterized by an element of $\Pi_{n-1}(G)$ ²⁰ which, for the case of interest is $\Pi_1(U(1)) = \Pi_3(SU(2)) \simeq \mathbb{Z}$.

Then, if the gauge field configuration A_μ defines a connection on a principal fibre bundle $P(S^n, G)$, we can compute its topological characterization from the expression

$$q = \int \mathcal{D}, \quad (3.1)$$

where the topological density $\mathcal{D}$ is given by

$$\mathcal{D} = \frac{1}{4\pi} F_{\mu\nu} \epsilon^{\mu\nu} d^2x, \quad n=2, \quad G=U(1), \quad (3.2)$$

$$\mathcal{D} = \frac{1}{64\pi^2} \text{tr}(F_{\mu\nu} F_{\alpha\beta} \epsilon^{\mu\nu\alpha\beta}), \quad n=4, \quad G=SU(2). \quad (3.3)$$

Even for A_μ with GI singularities [that is, not defining a principal fibre bundle $P(S^n, G)$] the quantity given by expression (3.1) can be considered as a topological invariant associated to A_μ , since its value does not change when A_μ is locally deformed. Of course, this quantity does not correspond in general to the topological characterization of a fibre bundle $P(S^n, G)$.

If A_μ has a singularity at $x = x_0$ ($x_0 \neq \infty$), applying Stokes' theorem one obtains

$$q = \lim_{\epsilon \rightarrow 0} \int_{|x-x_0|=\epsilon} K, \quad (3.4)$$

where

$$K = K_\mu dx^\mu, \quad n=2$$

$$K_\mu = \frac{1}{2\pi} A_\mu, \quad G=U(1); \quad (3.5)$$

$$K = K_\mu n^\mu d^3x, \quad n=4, \quad G=SU(2),$$

$$K_\mu = \frac{1}{32\pi^2} \epsilon_{\mu\nu\lambda\sigma} [A_\nu (F_{\lambda\sigma} - \frac{2}{3} A_\lambda A_\sigma)]. \quad (3.6)$$

We can interpretate expression (3.4) saying that the topological charge q is concentrated at $x = x_0$. It is then natural to consider the r.h.s. of Eq. (3.4) as giving the *local topological charge* of A_μ at x_0 . (Of course this concept is not gauge invariant. For instance at a GM singularity the local

charge can be displaced from x_0 to another point by a singular gauge transformation.)

If A_μ has several singularities, q will be given by the sum of all local topological charges.

We will now establish the following proposition.

Proposition: Let x_0 be an isolated singularity of A_μ . If the local topological charge

$$q = \lim_{\epsilon \rightarrow 0} \int_{|x-x_0|=\epsilon} K \quad (3.7)$$

is noninteger ($q \notin \mathbb{Z}$), then x_0 is a gauge-invariant singularity. (If $x_0 = \infty$, see Ref. 19.)

Demonstration: Suppose that x_0 is a GM singularity of A_μ and let us call Ψ a gauge transformation, regular on $\mathcal{U} - \{x_0\}$ such that the gauge-transformed field A'_μ is regular at x_0 . (Here $\mathcal{U}$ is a neighborhood of x_0 not containing any other singularity of A_μ .)

Let us take a function f , smooth $f: S^n \rightarrow \mathbb{R}$ with

$$\begin{aligned} f &\equiv 1 && \text{on } \mathcal{U}, \\ f &\equiv 0 && \text{on a neighborhood of all other} \\ &&& \text{singularities of } A_\mu \text{ on } S^n \end{aligned} \quad (3.8)$$

Then

$$\tilde{A}_\mu = f A_\mu \quad (3.9)$$

defines a configuration with a singularity at $x = x_0$. The local charge of $\tilde{A}_\mu$ is of course equal to that of A_μ . This gauge field $\tilde{A}_\mu$ defines a connection on the principal fibre bundle $\tilde{P}(S^n, G)$ constructed from the transition function Ψ . This means that the local charge of $\tilde{A}_\mu$, Eq. (3.7), coincides with the topological charge of $\tilde{P}(S^n, G)$. But this charge must be an integer, in contradiction with the proposition's hypothesis. Then, if the local charge of A_μ at x_0 is not an integer, the singularity is gauge invariant, Q.E.D.

In the examples of configurations with noninteger topological charges [Eqs. (2.11) and (2.22)], we have seen that it was not possible to find a gauge transformation in order to displace the singularities. The preceding proposition generalizes this result giving a geometrical interpretation of this fact.

We can apply the proposition to the configuration discussed by Marino and Swieca,¹⁶ already mentioned in Sec. II, which has a noninteger local topological charge (at ∞). Then, the singularity originated by the ill definiteness of A_μ at the point at infinity is not gauge removable.

Finally, we want to comment on the relation between our results and a discussion given by Crewther in Refs. 7 and 8. In Sec. 2 of Ref. 7 there are examples [related to those of Ref. 16] of configurations which cannot be compactified. In the context of our discussion, this corresponds to configurations which do not define a connection on a principal fibre bundle with base S^n .

Then, from our results we see that the configurations that are important from the point of view of Ref. 8 must have GI singularities.

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